Seminar 14 - GR 312 CD - 27.01.2022

Sisteme liniare omogene cu coef constanți

$$\begin{cases} x'(t) = a\cdot x(t) + b\cdot y(t) \\ y'(t) = c\cdot x(t) + d\cdot y(t) \end{cases}$$

a, b, c, d cunoscute.

$x(\cdot); y(\cdot) = ?$

↗ metoda valorilor și vectorilor proprii

↘ metoda reducerii la o ec de ord 2.

Ex 1

$$\begin{cases} x'(t) + 2x(t) - y(t) = 0 & (1) \\ y'(t) + 4x(t) - 3y(t) = 0 & (2) \end{cases} \quad \forall t \in I \Rightarrow$$

o ec omog de ord 2: $a\cdot x''(t) + bx' + cx = 0$ ⇓ $x(t)$ ...

$$(1) \Rightarrow \boxed{y(t) = x'(t) + 2x(t)} \quad \Big| (\,)' \Big] \overset{(2)}{\Rightarrow}$$

$$y'(t) = x''(t) + 2x'(t)$$

$$\Rightarrow (x'' + 2x') + 4x - 3(x' + 2x) = 0.$$

$$\boxed{x'' - x' - 2x = 0}$$

$$\lambda^2 - \lambda - 2 = 0 \Rightarrow \Delta = 1 + 8 = 9.$$

$$\lambda_{1,2} = \frac{1 \pm 3}{2} = \begin{cases} 2 \\ -1 \end{cases}.$$

$$x(t) = c_1 e^{2t} + c_2 e^{-t}.$$

$$y(t) = x'(t) + 2x(t) = \left(2c_1 e^{2t} - c_2 e^{-t}\right) + 2\left(c_1 e^{2t} + c_2 e^{-t}\right)$$

$$\boxed{y(t) = 4c_1 e^{2t} + c_2 e^{-t}.}$$

Metoda valorilor și vectorilor proprii

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix}; \quad \vec{\varphi}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} \Rightarrow \vec{\varphi}'(t) = \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}.$$

$$\boxed{\vec{\varphi}'(t) = A\cdot\vec{\varphi}(t)} \quad (*)$$

? $\vec{\varphi}(t) = e^{\lambda t}\cdot\vec{v}$ sol pt $(*) \Rightarrow \vec{\varphi}'(t) = \lambda e^{\lambda t}\vec{v}$

$\vec{\varphi}'(t) = Ae^{\lambda t}\vec{v}$] $\Rightarrow \boxed{A\vec{v} = \lambda\vec{v}}$

⇓

λ val pr.

$\vec{v}$ vect pr.

$X' = A \cdot X$ (*).

1) val. proprii pt matr. $A \to \lambda_1, \lambda_2 \in \mathbb{R}$; $\lambda_1 \neq \lambda_2$ $\Rightarrow e^{\lambda_1 t}, e^{\lambda_2 t}$ sol. pt (*)

2) vect. proprii v_1, v_2.

3) $\boxed{\vec{X}(t) = c_1 \cdot e^{\lambda_1 t}\vec{v_1} + c_2 e^{\lambda_2 t}\vec{v_2}}$

Ex 2 $\begin{cases} x' = x(t) + y(t) \\ y' = 4x(t) + y(t) \end{cases}$

$\Gamma \quad \begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow X' = \underbrace{\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}}_{A} \cdot X$; $X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$.

i) $P_A(\lambda) = \det(A - \lambda I_2) \Rightarrow \lambda_1 = -1$; $\lambda_2 = 3$.

ii) $\lambda_1 = -1 \Rightarrow v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$; $\boxed{\lambda_2 = 3} \Rightarrow ? \; v = \begin{pmatrix} a \\ b \end{pmatrix}$ i. $Av = \lambda_2 v$.

$\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 3\begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow a + b = 3a \Rightarrow b = 2a \Rightarrow v = \begin{pmatrix} a \\ 2a \end{pmatrix} \Rightarrow \boxed{v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}}$

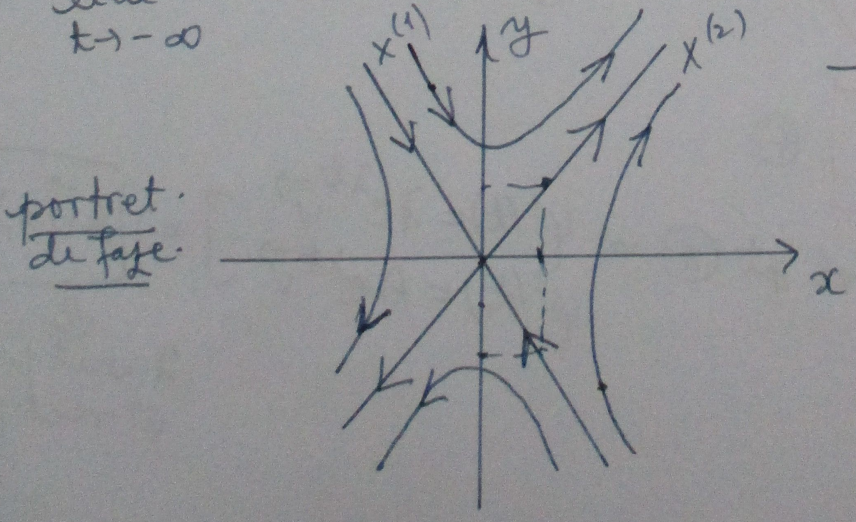
iii) $X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 \cdot e^{\lambda_1 t}\cdot v_1 + c_2 e^{\lambda_2 t} v_2 =$

$= c_1 e^{-t} \underbrace{\begin{pmatrix} 1 \\ -2 \end{pmatrix}}_{x^{(1)}} + c_2 \cdot e^{3t} \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{x^{(2)}} = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$.

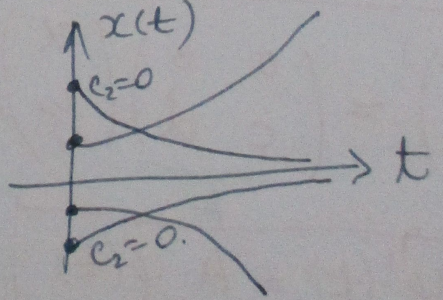
$\begin{cases} x(t) = c_1 e^{-t} + c_2 e^{3t} \\ y(t) = -2c_1 e^{-t} + 2c_2 e^{3t} \end{cases}$

$\lim_{t \to \infty} x(t) = \pm\infty$ ($c_2 \neq 0$. termen dom e^{3t})

$\lim_{t \to -\infty} x(t) = \pm\infty$ (termen dom e^{-t}).



portret de faze.



$X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$.

$X^{(2)} = e^{3t}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$; $X^{(1)} = e^{-t}\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Ex 2 $X' = \underbrace{\begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix}}_{A} X$; ? nod gen ; gf x_1, x_2.

a) $P_A(\lambda) = \det(A - \lambda I_2) \Rightarrow \lambda_1 = -1 \Rightarrow V_1 = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} \Rightarrow X^{(1)}(t) = e^{-t}\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$

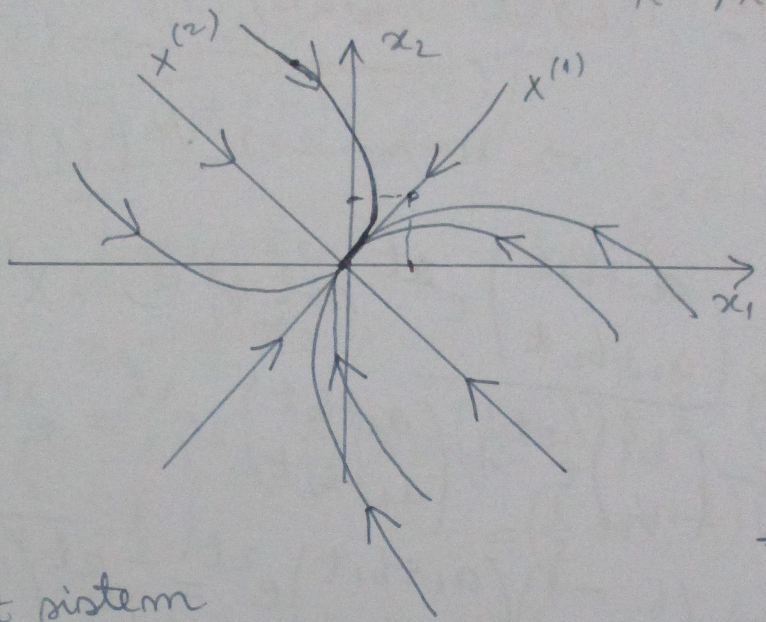
$\lambda_2 = -4 \Rightarrow V_2 = \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} \Rightarrow X^{(2)}(t) = e^{-4t}\begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix}$

$$X(t) = c_1 \underbrace{e^{-t}\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}}_{X^{(1)}} + c_2 \underbrace{e^{-4t}\begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix}}_{X^{(2)}}.$$

$\det(X^{(1)} | X^{(2)}) \neq 0 ; \forall t$.

$\Rightarrow X^{(1)}; X^{(2)}$ lin indep.

$t \to \infty \downarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$



$t \to \infty$.

term dominant $\Rightarrow X^{(1)}$

$X(t)$ se apropie de origine tang la dr $x_2 = \sqrt{2} \cdot x_1$

$t \to -\infty$.

$\Rightarrow$ term dom $X^{(2)}$.

0 = nod pt sistem stabil. $\Rightarrow$ valori proprii de acelasi semn.

Obs: $\lambda_1 > 0 ; \lambda_2 > 0 \Rightarrow$ sensul de fugire $\to \pm\infty$.

Ex 3

$$\begin{cases} x_1' = x_2 + x_3 \\ x_2' = x_1 + x_3 \\ x_3' = x_1 + x_2 \end{cases} \Rightarrow \boxed{X' = \underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}}_{A} X}; \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$P_A(\lambda) \Rightarrow \lambda_1 = 2 ; \lambda_2 = \lambda_3 = -1 \Rightarrow V_1 = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{\lambda_1}$; $V_2 = \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}_{\lambda_2 = \lambda_3}$; $V_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.

$$X(t) = c_1 \underbrace{e^{2t} V_1}_{X^{(1)}} + c_2 \underbrace{e^{-t} \cdot V_2}_{X^{(2)}} + c_3 \underbrace{e^{-t} V_3}_{X^3}.$$

V_2, V_3 lin ind

$\det[X^{(1)} | X^{(2)} | X^{(3)}] \neq 0 \ \forall t$

Valori proprii repetați

$X' = A \cdot X$; $P_A(\lambda) = 0$; $\lambda_1 = \lambda_2 = r$ val. proprie

$n=2$ → $\begin{cases} v_1, v_2 \text{ vectori proprii liniar ind. ; } \dim V_\lambda = 2 \\ \vec{X}(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2 \end{cases}$ $n>2$

$\vec{v}$ vector propriu ; $\dim V_\lambda = 1$.

ec de ord II → e^{rt} ; $te^{rt} \Rightarrow Ae^{rt} + Bte^{rt}$; $A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, $B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} e^{rt} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} te^{rt} = \begin{pmatrix} a_1 + b_1 t \\ a_2 + b_2 t \end{pmatrix} e^{rt}$$

Ex 4

(*) $\begin{cases} x_1' = x_1 - x_2 \\ x_2' = x_1 + 3x_2 \end{cases} \Rightarrow \lambda_1 = \lambda_2 = 2 \Rightarrow V = \begin{pmatrix} -1 \\ +1 \end{pmatrix}$ vect. pr. X

$$X(t) = \begin{pmatrix} a_1 + b_1 t \\ a_2 + b_2 t \end{pmatrix} e^{2t} \text{ sol pt (*)} \Rightarrow X'(t) = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$X'(t) = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} e^{2t} + \begin{pmatrix} a_1 + b_1 t \\ a_2 + b_2 t \end{pmatrix} 2e^{2t} = e^{2t} \begin{pmatrix} b_1 + 2a_1 + 2b_1 t \\ b_2 + 2a_2 + 2b_2 t \end{pmatrix}$$

$$X'(t) = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} a_1 + b_1 t \\ a_2 + b_2 t \end{pmatrix} e^{2t} = e^{2t} \begin{pmatrix} a_1 + b_1 t + a_2 - b_2 t \\ a_1 + b_1 t + 3(a_2 + b_2 t) \end{pmatrix}$$

$$\begin{cases} b_1 + 2a_1 = a_1 - a_2 \Rightarrow a_1 = -a_2 - b_1 \\ 2b_1 = b_1 - b_2 \Rightarrow b_1 = -b_2 \text{ ; } b_1 = \beta \Rightarrow b_2 = -\beta \end{cases} \Rightarrow \begin{cases} a_1 = \alpha \\ b_2 = -\beta \end{cases}$$

$\Rightarrow a_2 = -\alpha - \beta$

$$X(t) = \begin{pmatrix} \alpha + \beta t \\ -\alpha - \beta - \beta t \end{pmatrix} e^{2t} = -\alpha \cdot \begin{pmatrix} -1 \\ +1 \end{pmatrix} e^{2t} + \beta \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t} + \beta \begin{pmatrix} -1 \\ +1 \end{pmatrix} te^{2t}$$

(!)

Valori proprii complexe

$X' = A \cdot X$; $A \in M_2(\mathbb{R})$ $P_A(\lambda) = 0 \to \lambda$ val proprie ; $\vec{v}$ vector propriu.

$\Rightarrow e^{\lambda t} \cdot \vec{v}$ sol pt sist.

$\begin{cases} \lambda_1 = a + ib \to v_1 \\ \lambda_2 = a - ib \to v_2 \end{cases}$; $\overline{v_2} = v_1$ vect. conjugati.

v_1 vect pr pt $\lambda_1 \Rightarrow Av_1 = \lambda_1 v_1 \to \overline{Av_1} = \overline{\lambda_1 v_1} \to \overline{A} \cdot \overline{v_1} = \overline{\lambda_1} \overline{v_1}$

$A\overline{v_1} = \underbrace{\overline{\lambda_1}}_{\lambda_2} \cdot \overline{v_1} \Rightarrow A\overline{v_1} = \lambda_2 \overline{v_1} \Rightarrow \underline{v_2 = \overline{v_1}}$.

$X^{(1)}(t) = e^{\lambda_1 t} v_1 \Rightarrow u(t) = \mathrm{Re}\, X^{(1)}$; $v(t) = \mathrm{Im}\, X^{(1)}$ sol. reale liniind.

$$\boxed{X(t) = c_1 \cdot u(t) + c_2 v(t)}$$

EX 5. $\begin{cases} x_1' = -\frac{1}{2}x_1 - x_2 \\ x_2' = x_1 - \frac{1}{2}x_2 \end{cases}$; ? x_1, x_2 ; Gf x_1 vs. x_2.

$A = \begin{pmatrix} -\frac{1}{2} & -1 \\ 1 & -\frac{1}{2} \end{pmatrix} \Rightarrow P_A(\lambda) = \left(\frac{1}{2} + \lambda\right)^2 + 1 = \left(\frac{1}{2} + \lambda\right)^2 - i^2 =$

$= \left(\frac{1}{2} + \lambda + i\right)\left(\frac{1}{2} + \lambda - i\right)$.

$\begin{cases} \lambda_1 = -\frac{1}{2} + i \\ \lambda_2 = -\frac{1}{2} - i \end{cases} \Rightarrow v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$; $v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$.

$X^{(1)}(t) = e^{\lambda_1 t} v_1 = e^{\left(-\frac{1}{2} + i\right)t} v_1 = e^{-\frac{t}{2}} \cdot e^{it} v_1$

$X^{(1)}(t) = e^{-t/2}(\cos t + i \sin t) \begin{pmatrix} 1 \\ i \end{pmatrix}$.

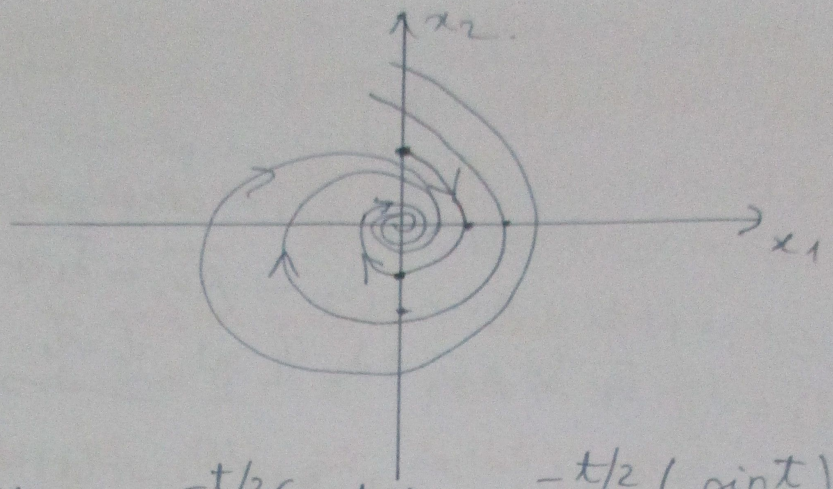
$X^{(1)}(t) = e^{-t/2} \begin{pmatrix} \cos t + i \sin t \\ -\sin t + i \cos t \end{pmatrix} \Rightarrow u(t) = e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$

$v(t) = e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$.

$$\boxed{X(t) = c_1 e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}.}$$

$\Rightarrow$ liniind. $\det[u|v] \neq 0$

lin ind. $\Leftarrow \det[u|v] = \begin{vmatrix} e^{-t/2}\cos t & e^{-t/2}\sin t \\ -e^{-t/2}\sin t & e^{-t/2}\cos t \end{vmatrix} = e^{-t}(\cos^2 t + \sin^2 t) = e^{-t} \neq 0$ ∀ t.



$$\vec{X}(t) = c_1 e^{-t/2}\underbrace{\begin{pmatrix}\cos t\\ -\sin t\end{pmatrix}}_{u(t)} + c_2 e^{-t/2}\underbrace{\begin{pmatrix}\sin t\\ \cos t\end{pmatrix}}_{v(t)} \xrightarrow{t\to\infty} \begin{pmatrix}0\\0\end{pmatrix}.$$

$$v(t) = e^{-t/2}\begin{pmatrix}\sin t\\ \cos t\end{pmatrix};\ v(0) = e^{-0/2}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}.$$

$$v\left(\frac{\pi}{2}\right) = e^{-\pi/4}\cdot\begin{pmatrix}1\\0\end{pmatrix};\ v(\pi) = e^{-\pi/2}\cdot\begin{pmatrix}0\\-1\end{pmatrix};\ v\left(\frac{3\pi}{2}\right) = e^{-\frac{3\pi}{4}}\begin{pmatrix}-1\\0\end{pmatrix}.$$

Sisteme neomogene. Met. var const.

$$X' = A\cdot X + B(t).$$

I) $X' = A\cdot X \Rightarrow X^{(1)}; X^{(2)} \Rightarrow X_0(t) = c_1 X^{(1)} + c_2 X^{(2)}.$

II) $X' = AX + B(t) \Rightarrow X(t) = c_1(t) X^{(1)} + c_2(t) X^{(2)}.$

unde c_1', c_2' verifică $\boxed{c_1'\cdot X^{(1)} + c_2'\cdot X^{(2)} = B(t).}$

Obs: $\boxed{A = PDP^{-1}}. \Rightarrow \begin{pmatrix}x_0\\y_0\end{pmatrix} = e^{At}\cdot\begin{pmatrix}c_1\\c_2\end{pmatrix}$ și $e^{At} = Pe^{Dt}P^{-1}.$

$$X_0(t) = M\cdot\begin{pmatrix}c_1\\c_2\end{pmatrix}.$$

$$\boxed{M\cdot\begin{pmatrix}c_1'\\c_2'\end{pmatrix} = B(t).} \Rightarrow \begin{pmatrix}c_1'\\c_2'\end{pmatrix} = \dots$$

Ex 6 $\begin{cases} x_1' = 2x_1 - x_2 + e^t \\ x_2' = -2x_1 + 3x_2 + e^{-t} \end{cases}$

$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$; $A = \begin{pmatrix} 2 & -1 \\ -2 & 3 \end{pmatrix}$; $B(t) = \begin{pmatrix} e^t \\ e^{-t} \end{pmatrix} \Rightarrow X' = AX + B(t)$

I) $X' = A \cdot X \Rightarrow P_A(\lambda) = \begin{vmatrix} 2-\lambda & -1 \\ -2 & 3-\lambda \end{vmatrix} \Rightarrow \lambda_1 = 1 \Rightarrow V_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\lambda_2 = 4 \Rightarrow V_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$$X_0(t) = C_1 e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + C_2 e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$\qquad\qquad X^{(1)} \qquad\qquad X^{(2)}$

II) $X' = AX + B(t) \Rightarrow X(t) = C_1(t) e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + C_2(t) e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

unde C_1', C_2' verifică $C_1' \cdot X^{(1)} + C_2' \cdot X^{(2)} = B(t)$.

$$C_1' \cdot e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + C_2' \cdot e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} e^t \\ e^{-t} \end{pmatrix} \Rightarrow$$

$$\begin{cases} C_1' e^t + C_2' e^{4t} \cdot (-1) = e^t \quad | \cdot 2 \\ 3C_1' e^t + 2C_2' e^{4t} = e^{-t} \end{cases} \overset{(+)}{\Rightarrow} 5C_1' e^t = 2e^t + e^{-t}.$$

$$(*) \quad C_1' = \frac{2}{5} + \frac{1}{5} e^{-2t} \Rightarrow C_1(t) = \frac{2}{5} t - \frac{1}{10} e^{-2t} + k_1.$$

$$(2) \Rightarrow C_2' \cdot e^{4t} = C_1' \cdot e^t - e^t = e^t (C_1' - 1) \overset{(*)}{=} e^t \left(-\frac{3}{5} + \frac{1}{5} e^{-2t} \right)$$

$$\Rightarrow C_2' = e^{-3t} \left(-\frac{3}{5} + \frac{1}{5} e^{-2t} \right) = -\frac{3}{5} e^{-3t} + \frac{1}{5} e^{-5t}$$

$$C_2(t) = \frac{1}{5} e^{-3t} - \frac{1}{25} e^{-5t} + k_2.$$

$$X(t) = \left(\frac{2}{5} t - \frac{1}{10} e^{-2t} + k_1 \right) e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \left(\frac{1}{5} e^{-3t} - \frac{1}{25} e^{-5t} + k_2 \right) e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

$$= X_0(t) + X_p(t).$$

$X_p = te^t \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + e^{-t} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} + e^t \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$